Measuring Data Accuracy: Progress and Perils
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MEASURING DATA ACCURACY: PROGRESS AND NEXT STEPS

ABSTRACT

Perhaps no management mantra is repeated more often than "you can't manage what you don't measure." The mantra is no less true for data quality, especially *data accuracy*. Measurement is always difficult, but data accuracy presents complications that have no analogue in physical measurement. This paper summarizes these complications. The result is that, with few exceptions, data accuracy measurements are surrogates. These difficulties aside, data accuracy may be measured in dozens of ways. We propose a *data accuracy measurement framework* based on four factors for understanding candidate measurements. As an example, one factor is "where the measurement is taken." Each measurement method has strengths and weaknesses, so practitioners have difficult choices to make. Therefore, we introduce the concept of a *measurement system* as a means to select a suitable data accuracy measurement based on an organization's particular circumstances. Finally, we illustrate the use of the measurement system in three common data quality situations.

1. Introduction and Summary

Perhaps no management mantra is quoted more often than "You can't manage what you don't measure." The mantra is no less true for data quality. Herein we focus on data accuracy, probably the most basic dimension of data quality. Without pertinent measurements of the accuracy of data created within an information chain or provided by a supplier, it is simply too difficult to make and sustain improvements. And it is much more difficult for data customers,

those who use data in operations and decision making, to trust the data. Instead, managers are reduced to "management by anecdote" or worse, "management by unsupported opinion."

Of course if data were just "the stuff stored away in databases," this situation would be bad enough. But four interrelated trends make improved data accuracy, and hence its measurement, more urgent. First, inaccurate data are increasingly costly. They lie at the root of many of the most important issues of the day—from lack of trust in financial markets, to contested elections, to added difficulties in providing homeland security. Most data quality issues are, fortunately, more mundane. But they are harmful nonetheless, increasing expense, angering customers, and compromising decision making. Second, the way organizations work is changing. Supply chain management is but one example. Collectively, these changes are replacing people, formerly the last line of defense against inaccurate data, with data exchanges and software. The result is that more bad data slip through. Third, the sheer quantities of data created and stored are growing exponentially. If nothing else, more data mean more data are unfit for use! Finally, the Internet contributes to each of these trends in ways that are not fully understood. The Internet does, for example, make more data available to more people, some of whom may be more easily victimized by inaccurate data.

Measurement is never easy. And in many domains a sophisticated body of theory and technique, collectively called *metrology,* has evolved (see for examples, [2] and [12]). Many principles apply to data accuracy. But there are special difficulties that stem from the nature of data. For example, data have no physical properties such as "length" or "viscosity," and hence there are no physical properties to measure. Indeed, the most basic question, "how accurate are the data?" presents complications that have no analogue in physical measurement. We discuss these difficulties in Section 2.

Section 3 provides a short description of the two predominant approaches to data quality. The so-called *first-generation* approach aims to find and fix errors, while the *second-generation* approach aims to prevent errors at their sources. Since the second-generation approach consistently yields higher-quality data at lower cost, we favor measurements that support it.

The difficulties discussed in Section 2 aside, progress has, of course, been made on many fronts. There are tools to evaluate whether data pass business rules, customer-satisfaction surveys, estimates of the cost-of-poor-data-quality, and sophisticated data tracking techniques. In Section 4 we propose a *data accuracy measurement framework* for understanding the various measurement techniques based on choices made regarding four factors:

- Where the measurements are taken,
- The measurement device,
- What data are included, and
- How results are summarized.

We also briefly describe possible choices for each factor.

Each of the many ways to measure data accuracy has strengths and weaknesses and there is no one best way to measure data accuracy that fits all circumstances. Thus in Section 5 we introduce the concept of a *measurement system* as a means of selecting an accuracy measurement based on an organization's particular circumstances. By a measurement system we mean the totality of measurement effort. The concept is a broad one. For example, the abilities of management to use results to manage and improve data quality are explicitly included.

In Sections 5 and 6, three measurement methods that have proven especially effective in common data quality situations are described in more detail. These methods still suffer the

difficulties described in Section 2. So in Section 7 we offer advice to practitioners on how to cope.

2. Data, Data Quality, and Accuracy Defined

Philosophers, computer scientists, and information systems engineers have debated just what "data" (and other Information Age assets such as "information," knowledge," and "wisdom") are for decades. We've selected the definition used herein for a variety of theoretical and practical reasons (see [4] and [13]); the most important being that it suits the "second-generation" approach to data quality management. Thus, data (or a data collection, data set, etc.) consist of two related components: a *data model* and *data values*. Data models are abstractions of the real world that define what the data are all about. Models define entity classes, the collections of things (physical or ideal) of interest; and attributes and relationships, which specify pertinent features of the entities. As an example, the reader is an entity. His employer is interested in the entity class "EMPLOYEES" and attributes such as NAME, DEPARTMENT, SALARY, and MANAGER. The Internal Revenue Service is also interested in the reader, as a "TAXPAYER." It is interested in many of the attributes above and others, such as INTEREST INCOME.

Within this framework, each attribute must have a well-defined domain of values that are permissible under the model. For example, a list of an organization's departments defines the domain of possible values for DEPARTMENT. Data values are to be drawn from the domain and assigned to attributes and/or relationships for specific entities. A single datum is thus a triple:

Datum = <entity, attribute, value>

An example datum is:

<Thomas C. Redman, DEPARTMENT = 11201>

Note that the data value assigned to a given <entity, attribute> pair may be incorrect. Further, several values for the same <entity, attribute> pair may be contained in different databases and the respective values may disagree.

Note also that data, as defined in this manner, are abstract. *Data records* are the physical manifestations of data, stored in paper files, on spreadsheets, and in electronic databases. It is data records that customers, those who use the data, actually see.

It is clear that any measurement of data will present complications. First, we noted above that data are intangible—they have no physical properties and hence there are no physical properties to measure. There is nothing akin to length, viscosity, impurities in parts per million, impedance, or other physical dimensions. A practical consequence is that you cannot tell by direct examination whether a data value is correct. All measurements of data accuracy must, of necessity, make reference to human knowledge, other data, or the real world. Reference to the real world is almost always costly. Other measurements are indirect measurements or surrogates, complicating interpretation.

A somewhat more subtle distinction is that most useful data are novel or unique. For example, one could define an attribute GENUS/SPECIES for employees. But all values would be the same, *Homo sapiens*. These data are not interesting. Interesting data take on distinct values and information chains that create new data values must accommodate this novelty. This stands in contrast with most manufacturing processes, where one strives for uniformity and standard measurements can readily be applied. But there are no such standards for data.

A further complication is that data values cannot be measured until they are recorded in some fashion, often in a database. But there is an inherent lag in updating a database. As an

example, consider a magazine's list of subscribers. When a subscriber moves, she may well inform the magazine of her new address. But the address data will be out of date, at least for a time. In some cases the lag may be long. Modern technologies, such as the Internet, may help reduce the lag. But it cannot be eliminated.

We next define *data quality* and *accuracy*. By definition, "data are of high quality if they are fit for their intended uses, by customers, in operations, decision making, and planning." [10] According to this definition, customers are the ultimate arbiters of quality. And customers can have a stunning range of needs for data. Some require "up-to-the-minute" data, others want data to be quite detailed, still others want data presented in an easy-to-understand format (see [3], [4], [10] and/or [14] for discussions of the various "dimensions of data quality"). Almost all customers want the data they use to be *correct*, meaning that data values agree with their real-world counterparts or with a standard that itself is assumed to be correct.

Accuracy measures the degree of correctness of a given collection of data. In some cases it is easy to define and measure accuracy. For example, the domain for the attribute SEX consists of two possible values: Male and Female. A data recording for this attribute is either correct or incorrect. Consider next an employee's YEAR OF BIRTH. Assuming the employee knows when he/she was born, it is simple enough to determine if the recording is correct. And if it is not correct, then it is also easy enough to quantify the inaccuracy, as the difference between the actual and recorded years. In practice, there are numerous situations such as these where measurement of data accuracy is straightforward.

But there are often complications. For example, the correct data value may be unknown. This is frequently the case with data gathered in the past—no corroborating evidence exists. Conversely, in some cases there may be more than one correct answer. As an example, many

foreign names have alternative spellings. Financial reporting presents another interesting twist. A reported datum, such as RECEIVABLES = $10,000,000 is itself an estimate. There may be plenty of corroborating evidence, but the true value, to the penny, may not be known. Finally the Presidential Election of the year 2000 provides a good example of the question "What constitutes a valid recording?" Is a "hanging chad" indicative of a vote or not?

Even when the correct value is known, it is not always clear how accuracy should be measured. In many circumstances it is easiest to simply note whether a data value is correct or incorrect. But doing so may ignore the consequence of the inaccuracy. If a letter is incorrectly addressed to APARTMENT = 3B instead of APARTMENT = 3C, the mail carrier may still deliver the letter. From one point of view, the letter was properly delivered. And if there is no consequence, why even bother to count the datum as incorrect? But there may be another point of view. Suppose the letter is for marketing purposes and the addressee reasons "I won't buy from someone who doesn't even know where I live." The consequences may be significant.

3. Second-Generation Data Quality Systems

While there are dozens of variants, all approaches to data quality fall into two categories. After Ishikawa [7], techniques that aim to find erred data values and correct them follow the so-called "first-generation" approach. These techniques, which are quite popular, include variations of the following:

- "direct inquiries," in which an organization asks its customers, suppliers, or others assumed to know correct values to check data records for errors and advise them of correct values.

- "database clean-up tools," in which software routines query databases to determine whether recorded data values lie within their specified domains and record discrepant data values for correction by specialists.
- "database bashing tools," which compare data values in two databases against one another. Data records that do not agree are referred to specialists for correction.

However implemented, first-generation techniques focus on data that have already been created. In contrast, "second-generation" techniques aim at preventing errors in data that will be created in the future. These techniques recognize that data are created within information chains, long cross-functional flows of data and information from data creator, through an organization, and to customers. For example, data about an employee are created when the employee joins the company, switches departments, gets a raise and/or promotion, and so forth (data models are created within the so-called "data modeling chain," so second-generation techniques apply to data models as well).

Figure 1 depicts the flow of data using the so-called "customer-supplier model" (see [10] for a fuller description of the customer-supplier model). To prevent future errors these techniques determine the root causes of errors, then modify or redesign the information chain process so it does not create similar errors in the future. Data quality improvement does not happen of its own accord. It is the result of diligent efforts to actively manage data creation. As a practical matter, this means that external data suppliers and internal business processes must be managed. Figures 2 and 3 present the data supplier and information chain management cycles respectively as means to do so. Note that measurement is the third and fourth steps of the respective cycles.

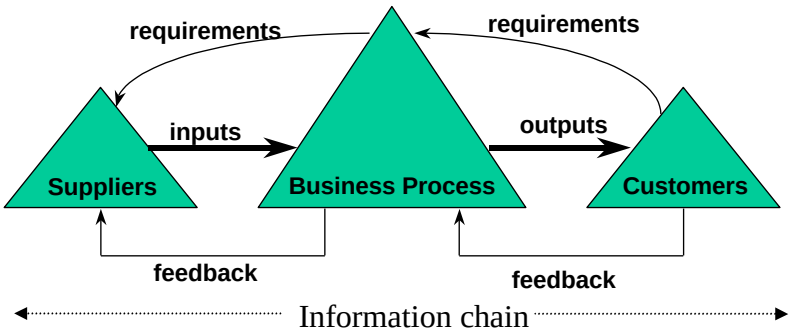


Figure 1: An information chain is a long, cross-organizational flow of data, from suppliers, through a given organization and its business processes, to customers, those who use the data. Existing data may be modified and new data created at each step.

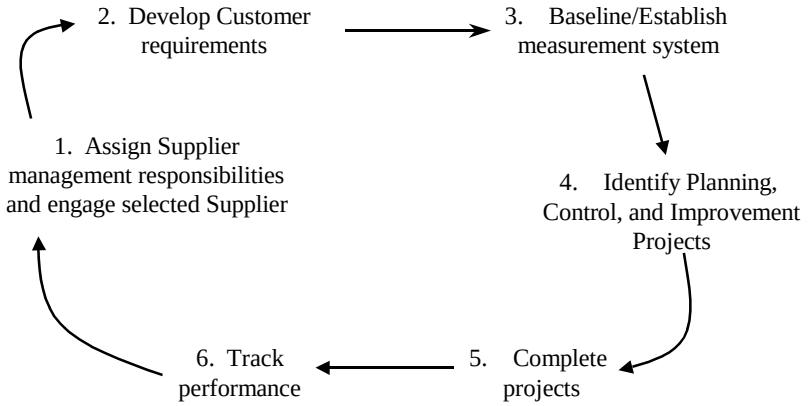


Figure 2: The Data Supplier Management Cycle is a structured approach for managing external data suppliers. Note that measurement is step 3.

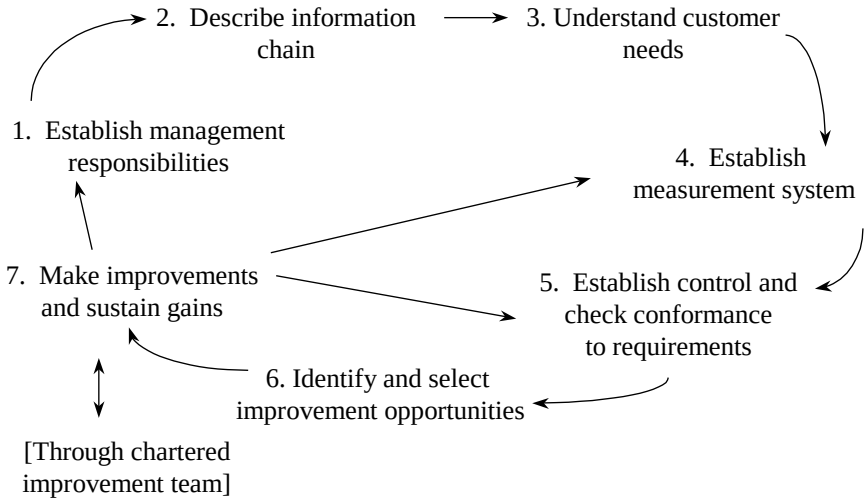


Figure 3: The Information Chain Management Cycle is a structured approach for managing data quality across a large information chain. Note that measurement is step 4.

4. A Data Accuracy Measurement Framework

The difficulties in measuring data accuracy noted in Section 2 aside, there are dozens of candidate measures of data accuracy. These candidates arise naturally as choices regarding four factors:

- Where measurements are taken,
- What data (i.e., attributes) are included,
- The measurement device, and
- The scale on which results are reported.

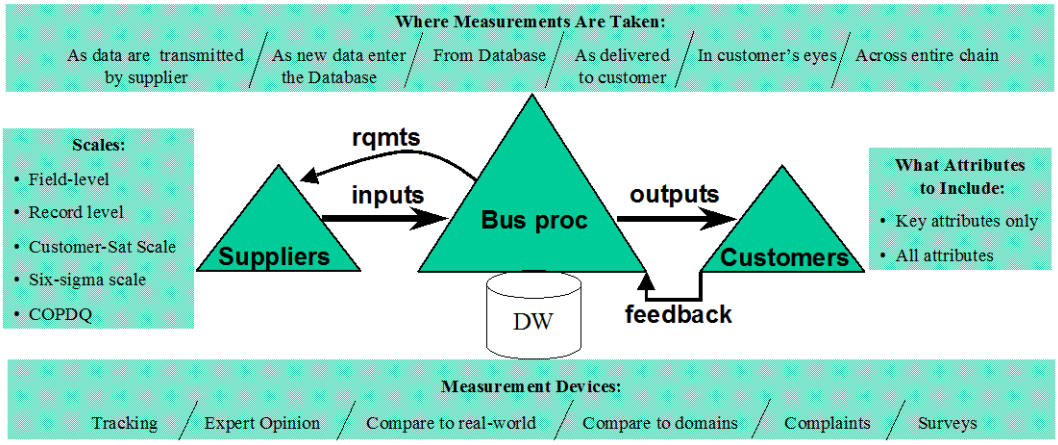


Figure 4: The Data Accuracy Measurement Framework. Note that the information chain/customer-supplier model has been modified somewhat. The four factors that define the framework surround it.

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Figure 4 presents a pictorial summary of the so-called Data Accuracy Measurement Framework. It is based on the customer-supplier model of Figure 1, modified to include a database (and simplified somewhat to minimize clutter). Surrounding the data flow are four factors and a number of choices for each factor (we cannot be certain that the lists of choices are complete). Across the top are choices for the "where" factor. The right-hand side lists choices of which data to include and the bottom lists measurement devices.

Finally, scales on which results may be reported are listed on the left-hand side. As described below, accuracy measurements can thus be made in (at least) six places, including all or a subset of the data, using (at least) six measurement devices, and reported on (at least) five scales.

First, consider where along the chain data accuracy can be measured.

- *As data are transmitted by a supplier.* If one is interested in improving supplier performance, it may well be appropriate to measure accuracy of the data as the supplier transmits them. It is often feasible for suppliers to make these accuracy measurements themselves.
- *As new data enter the database.* If one adopts a second-generation approach to data quality management, then one is most interested in how well the process of data creation works. It follows that attention should focus on newly-created data, or data that are "entering a database."
- *From the database.* The entire collection of data, as stored in a database, represents the resource available to the organization. So it may be appropriate to measure accuracy based on data in a database. Further, it is usually easier than making measurements anywhere else.
- *As the output data are delivered to the customer*. Since, in many cases, customers use only a very small fraction of data contained in a database, one may wish to make accuracy measurements as data are delivered to the customer.
- *As perceived by the customer.* Since "quality is in the eye of the customer," one could argue that data accuracy measurements involve customer perception.

- *Across the entire information chain.* Ideally, one desires effective (i.e., producing correct data values) and efficient (i.e., without rework and at low cost) information chains. This argues for measurement across the end-to-end information chain.

Second, consider which data are included in the measurement.

- *Key attributes/fields based on the most important uses of the data.* In most cases, some attributes (i.e., fields) are more important than others. Indeed, many records consist of dozens of attributes/fields, but relatively few are used. So it is often appropriate to concentrate on the most important.
- *All data (from a given database).* In other situations, such as data mining applications, it is difficult to speculate which data are most important. So all attributes/fields are included in the accuracy measurement.

Next consider measurement devices:

- *Data tracking throughout the information chain.* Data tracking provides a means of sampling data records as they first enter an information chain, tracking changes each step along the way, and applying business rules to identify inaccuracies. It is extremely powerful, especially at helping managers discover root causes of error [6].
- *Inspection by experts.* People who work with data are often extremely adept at spotting errors. Many seem to know business rules, even those they cannot articulate, intuitively. So error counts supplied by experts are often a good basis for measuring accuracy.
- *Comparison of data values to their real-world counterparts.* Since all other measurement devices are surrogates, comparing data values to their real-world

counterparts is, in many respects, a superior approach. Unfortunately, it is also time consuming, difficult, and usually expensive.

- *Comparison of data values to their domains of allowed values/business rules.* In most cases, it is straightforward to "count" data values that fall outside their domains of allowed values. The task is facilitated by software and many good tools are available. Importantly, the notion of a domain of allowed values extends to any number of attributes. For example, neither AREA CODE = 212 nor ZIP CODE = 90210 fall outside of their respective domains of allowed values. But the combination, as attributes associated with a person's home, fall outside the bivariate domain for the two attributes. Domains of allowed values are often formulated as "business rules" and there are rich taxonomies of these rules (see [11]) and ways to discover them (see for examples, [8] and [9]).
- *Customer complaints:* If adequate feedback loops are in place, it may be appropriate to measure data accuracy by "counting" complaints of inaccuracies.
- *Customer surveys*: Customer surveys aim to elicit customer opinion of data accuracy.

Finally, consider the scales for reporting:

- *At the "field" (attribute) level:* Here, accuracy is defined as:

$$field\,level\,accuracy = \frac{number\,of\ \ fields\ judged\ "correct"}{number\,of\ \ fields tested}$$

  The field level scale is perhaps the most basic.

- *At the "record" level.* Here a record is considered erred if any field is erred and the accuracy is defined as:

$$record\,level\,accuracy = \frac{number\,of\ records\ judged\ "completely\,correct"}{number\,of\ records tested}$$

It is usually the case that more than one field is needed to conduct an operation or make a decision and, if even one field is erred, the record must be reworked. Measuring accuracy at the record level therefore better reflects data accuracy in the customer's eyes.

- *On a customer satisfaction scale:* Customer satisfaction is usually reported based on the percentages of customers who respond that they are "very satisfied," "satisfied," "dissatisfied," "extremely dissatisfied" scale.
- *On the six sigma scale:* In recent years many companies have implemented six sigma programs. These programs use "opportunity for error" as a basis for creating measurements that can be (loosely anyway) compared across domains, organizations, etc. Error rates are put on a common scale based on the normal distribution. Table 1 gives a few sigma values and their associated error rates. As applied to data quality, each field can be thought of as an "opportunity for error." The measurement is thus based on field level reporting, translated into the sigma scale.

Sigma Level	Error Rate
1	0.698
2	0.309
3	0.081
4	0.0062
5	0.0002
6	0.0000034
After [1]	

Table 1: Selected sigma values.

- *Cost-of-poor-data-quality:* As we noted earlier, some data errors are more costly than others. Thus in certain circumstances it may be appropriate to report accuracy in terms of associated costs of error. This approach has the advantages that leaders of

organizations better understand costs and it is often easier to motivate improvements when costs are high.

To define a data accuracy measurement, one selects from among the choices for each of the four factors. As an example, one can measure data accuracy

- as the data are delivered by a supplier,
- using (violations of) business rules as the measuring device,
- focusing on the most important attributes, and
- reporting results at the record level.

While not all combinations yield sensible accuracy measurements, it is clear that there are a very large number of candidate accuracy measurements. And there is no "best" approach for all situations. So we next consider how one should select from the candidates.

5. Measurement Systems

We use the term *measurement system* to refer, collectively, to all activities that make up measurement, from deciding what to measure, to specifying the measurement protocol, to presenting results, to using results to make improvements. The concept aims to put measurement into its proper organizational contexts as, for example, data supplier and information chain managers have different needs. The concept also aims to facilitate best-practice measurement protocols and presentation devices. By viewing "measurement" as a system, the concept recognizes that measurements can be no more effective than their weakest component. For example, sophisticated measurements that are so complex that management cannot act on them are of no value. The concept of a measurement system applies to all measurement, not just data accuracy. And importantly, "measurement data" are themselves data. So the principles of data quality management apply to measurement data as well.

Measurement systems are particularly helpful in deciding how to measure data accuracy. As depicted in Figure 5, measurement systems consist of five interrelated components. By working through the components one can rather quickly select an appropriate accuracy measurement. Brief descriptions of each component follow. At the same time, we give an example that shows how each component applies to data accuracy.

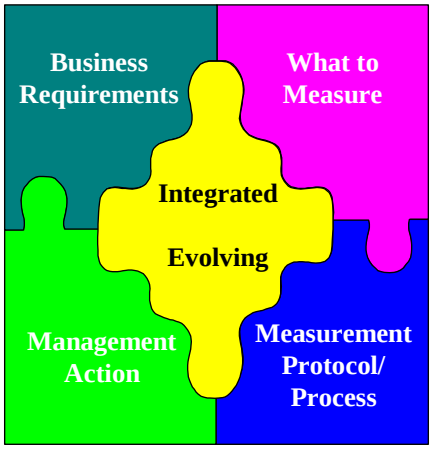


Figure 5: Effective Measurement Systems are made up of five inter-related components.

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The first component is called *business requirements*, though the term *measurement customers and their needs* is perhaps more descriptive. This component aims to make clear who will use the measurements and what for. For example, consider an organization that receives data each day from several suppliers, spot checks the data and corrects any obvious errors (or attempts to anyway), integrates the data into its own database, and makes them available to its customers. Later, if customers find errors, this organization must find and fix the erred values. Each of several suppliers provides much data, but ten attributes provided by one supplier are particularly important. In this case, the person responsible for the organization's "data supplier management" program is the primary measurement customer. First, she wants to get better data

from the most critical supplier so her organization can spend less time correcting data and more time creating new value for her customers. Since this supplier must undertake improvement projects to eliminate errors, it is important that its management be treated as a measurement customer also.

The second component involves specifying exactly what to measure. To do so, apply the data accuracy measurement framework. In the example:

- Where measurements are taken. Since the focus is on supplier quality, it is most appropriate that the supplier measure the accuracy of data it provides.
- What data to include. Since ten attributes are most important, the accuracy measurement should focus on them.
- Measurement device. While these data must agree with their real-world counterparts, data values that are simply absurd are the most troubling to this manager (they lead to customer mistrust). These can be identified by checking whether the value of each attribute falls within a pre-specified range (i.e., satisfies a business rule).
- Scale for reporting results: As used in this organization, a data record is unfit for use if even a single attribute is erred. So it is best to report results on the record level scale. At the same time, the data supplier manager wants to facilitate improvement, so error rates for each attribute are needed as well.

The third component is the measurement protocol, which specifies how measurements are taken, who will take them, where data are stored, how results are summarized and presented. If sampling is employed, the means of obtaining samples are part of this component. Finally, this component includes both specification and implementation of the protocol. In our example, the supplier agrees to:

STEP 1: Randomly select one hundred records a week.

STEP 2: Apply business rules to identify "absurd" values.

STEP 3: Summarize the record level error rate each week and provide a time-series plot and a Pareto chart of error rates for each attribute.

The fourth component is management action. In data quality management, data accuracy measures are used in two basic ways: to effect control and to identify root causes of error and eliminate them. In our example, the data supplier agrees that the initial focus will be on making improvements so that the record level error rate is no higher than 5%.

The final component is called Integration/Evolution. Appropriate measurements fit into their proper organizational contexts. In this case, the data supplier management program is rather basic. In time more attributes, more suppliers, and more data customers must be considered. These may well lead to more sophisticated measurements.

6. Evolving the Measurement of Data Accuracy

Progress in all disciplines and increasingly sophisticated measurement go hand-in-hand. The principle holds for data quality—as improvements make the data more accurate, increasingly sophisticated measurements are needed. The converse is also true. When an organization is starting its data quality program, it need not invest in sophisticated measurements. Rather simplistic measurements, used effectively, can add enormous value.

In Section 5, we stepped through one typical example. In this section we consider two more examples. The first is extremely basic and the second is quite sophisticated.

The first question many organizations ask is akin to "Are the data in the new data warehouse any good?" There is usually a good deal of contention around the question. A few vocal warehouse customers have convinced senior management that the data are poor on the

strength of a few examples. Warehouse staff, already overworked, claim "the data are as good as we can make them." Finally, few resources (time and people) are available to address the question.

Applying the measurement system and accuracy framework yields the following.

Customers and Their Needs: Everyone involved is a measurement customer and the most important need is to replace anecdote with fact.

What To Measure/Data Accuracy Measurement Framework:

- Where measurements are taken. The focus here is on a specific data warehouse, so it is the appropriate place to measure.
- What data to include. Again, owing to the shortage of time and money, it is probably best to focus on the attributes that key customers feel are inaccurate.
- Measurement device. Owing to the shortage of time and money, expert opinion is a good choice.
- Scale for reporting results. Both record level and attribute level summaries are needed.

Measurement Protocol: Since resources are tight, a relatively small sample of (say 100-500) records should be obtained (one can, of course, consult any text on statistical sampling to determine an exact sample size). To measure accuracy, the following procedure works well (refer also to Figure 6):

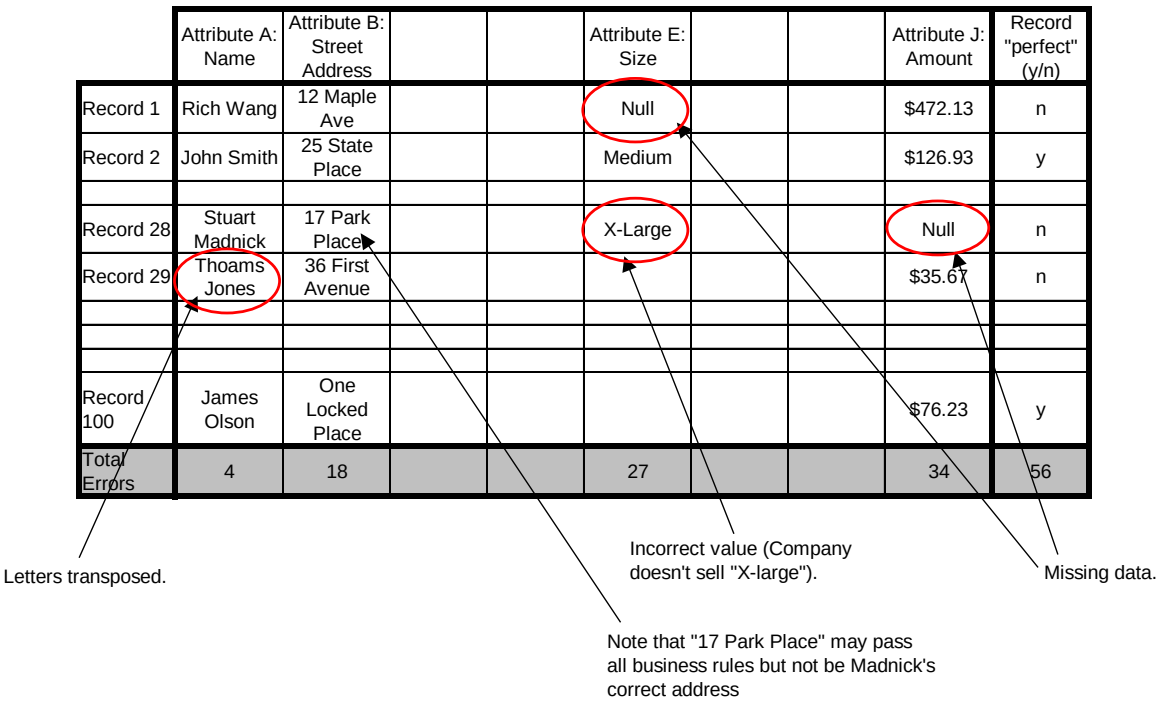


Figure 6: An example of a simple spreadsheet used by experts for estimating error rates

STEP 1: Obtain a sample of N data records, including J attributes. Lay these data out on a spreadsheet similar to Figure 6.

STEP 2: Have an expert review each record. Draw a red circle around each attribute that is obviously erred (according to the expert).

STEP 3: Summarize results along the margins of the spreadsheet. A record is erred if any field is erred. Count the number of errors for each attribute and for the entire record.

STEP 4: Summarize results.

Management Action: Management's principle goal is to determine whether the complaints of data warehouse customers are justified. If (say) thirty percent or more of the data records are erred, then management must initiate an aggressive data quality program. If only a couple of percent of data records are erred, then the contentions of data warehouse personnel are supported

and management will have to search elsewhere to determine the source(s) of customer dissatisfaction.

We next consider an organization that has implemented information chain management on its "Customer Order to Billing Chain." Data are created and cross several departmental lines, Order Entry to Credit Approval to Fulfillment to Inventory Management and, finally, to Billing. The chain generally works well, but there are enough customer complaints that the information chain owner wants to find out where errors occur and eliminate as many as possible.

Applying the measurement system and accuracy framework yields the following.

Customers and Their Needs: The owner of the information chain is the principle customer and his most important need is to identify where errors occur.

What To Measure/Data Accuracy Measurement Framework:

- Where measurements are taken. Clearly measurement should take place across the entire Order to Billing Chain.
- What data to include: Customers' complaints center on two problems: receiving the wrong merchandise and billing errors. So all attributes that impact merchandise delivery and billing should be included.
- Measurement device. Data tracking is the only appropriate measurement device.
- Scale for reporting results: Ultimately, management wants to reduce customer complaints. Since any erred attribute can cause a complaint, record level measurements are best.

Measurement Protocol: Since data tracking is expensive, a relatively small sample of N customer orders (N is typically 50-100) should be tracked each week. Note that management is

not designing a one-time measurement, but rather an ongoing measurement program (see [6] for a fuller description of data tracking).

STEP 1: Each week, obtain a sample of N customer orders as they enter the Order to Billing Process.

STEP 2: Track the data associated with these orders as they wind their way through the process.

STEP 3: Review tracked records to identify errors.

STEP 4: Summarize results. P-charts and Pareto charts are especially effective. See Figure 7.

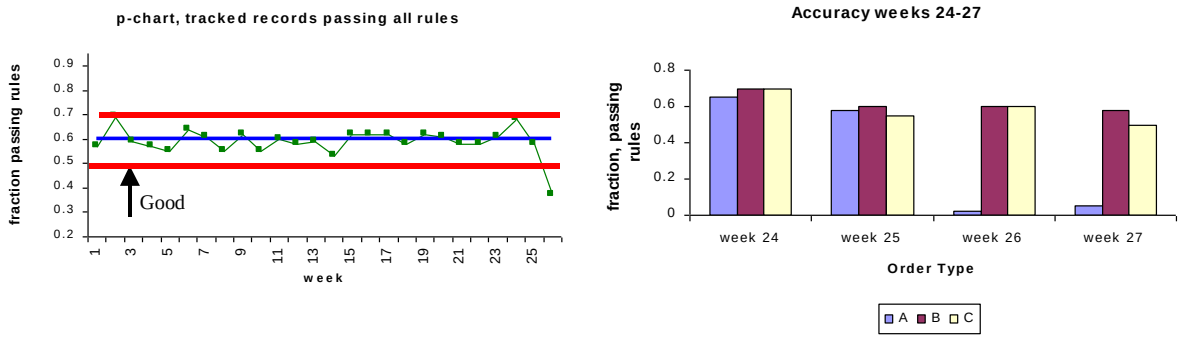


Figure 7: Together, so-called p-charts and Pareto charts are especially effective summaries for information chain managers.

Management Action: As noted, the information chain owner's principle goal is to determine where errors occur. Then, following the information chain management cycle, improvement projects can be chartered to eliminate them.

Importantly, these three accuracy measurements (one aimed at suppliers, the second at the data warehouse, the third at the entire information chain) complement one another. As noted, an organization may start with the data warehouse measurement described in this Section. Results could well highlight the need to measure supplier performance, and the measurement described in Section 5 can be implemented. Later, as the accuracy of supplier data improves, the organization may need to add data tracking.

7. Perils in Measuring Accuracy

With rare exceptions, most measurements of data accuracy have sidestepped the fundamental difficulties of measuring data accuracy raised in Section 2. This is not to impugn their utility—indeed many have proven quite useful. But sidestepping also leads to confusion and limitations.

Recall that accuracy, as defined, is based on the real world or on a standard that is itself assumed to be correct. Such standards are rare, and reference to the real world is time consuming, difficult, and expensive. So most accuracy measurements employ surrogate standards, usually informed humans (i.e., experts or customers) or business rules. The experts may indeed be well informed and the business rules may be excellent. But they are not the real world and accuracy measurements can be no better than the standards on which they are based. We use the term *measurement error* to mean the difference between a measurement based on standards and one based on the real world. It is almost always impossible to estimate measurement error. This is the most crucial issue that practitioners must bear in mind.

While there are no hard-and-fast rules, accuracy measurements based on business rules tend to underestimate the actual error rate (that is they are "biased"). This is because data values can pass business rules but still be incorrect. Experts both mistakenly identify correct values as incorrect and fail to spot incorrect values. A related point is that experts disagree, and business rules are themselves subject to interpretation.

Importantly, there are many situations in which measurement error, even if it is large, doesn't really matter. For example, suppose that, the accuracy rate, in the eyes of an expert, is low, say .5 (50%). Management must take action and immediately! Even if the expert is wrong

and the accuracy rate is somewhat higher (or much lower), management still must take immediate action!

The importance of measurement error grows as accuracy improves. When the true level of accuracy is 97%, a measurement error of a couple of percent can have a profound effect on management's best course of action. There is no simple prescription for dealing with this limitation, though we do offer the following advice to practitioners:

1. Define accuracy measurements very clearly. Make sure these definitions appear in all tables and plots used by others.
2. Strive to continuously improve measurements. Learn more sophisticated business rules and seek more diverse expert opinion and incorporate what is learned into the measurements.
3. Work closely with those who use the measurements to evolve and more sharply focus measurements. Most importantly, stop making measurements that are not proving useful and redirect the effort.
4. Experiment with different ways of presenting results.
5. From time to time, compare a sample of data values to the real world.

8. Concluding Remarks

In this paper we have presented an overview of the measurement of data accuracy. First, we have stressed the importance of doing so. We have described the twin concepts of a *data accuracy measurement framework* and a *measurement system* to help understand candidate accuracy measurements and to help select a good one based on the organization's particular circumstances. In doing so, we have enumerated where, along an information chain, measurements can be made, options for what data to include, potential measurement devices, and

scales for reporting results. We have shown, by example, how these concepts apply in situations that organizations and managers typically face. We have briefly illustrated important advances, including data tracking and the assimilation of powerful statistical tools. We have also illustrated simple yet effective measurements.

We have also described fundamental difficulties in measuring data accuracy and we have advised practitioners on how they can cope.

9. References

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